Geographic Patterns in County-Level PM2.5 Monitoring Data Across Wisconsin, 2024

Research question

The core question is where monitored county-level PM2.5 means were highest in Wisconsin during 2024 and whether those monitored counties showed meaningful spatial clustering.

Data and methods

The rebuild combines a 72-county TIGER/Line boundary layer, user-supplied Wisconsin PM2.5 summary files, and benchmark county means preserved in the rebuild prompt. County FIPS codes were standardized, unmonitored counties were retained in statewide mapping, and spatial diagnostics were run under both legacy Queen and KNN weights.

Main findings

Sixteen monitored counties appear in the legacy benchmark snapshot. Grant County is the highest monitored county and Monroe County the lowest. The descriptive correlation between PM2.5 and log population density is 0.604, but that relationship should be read cautiously given the small monitored sample.

Spatial-statistics caveat

The original Queen contiguity graph among monitored counties is disconnected and contains islands, which makes local cluster labels fragile. The KNN sensitivity analysis removes zero-neighbor counties and offers a more interpretable robustness check.

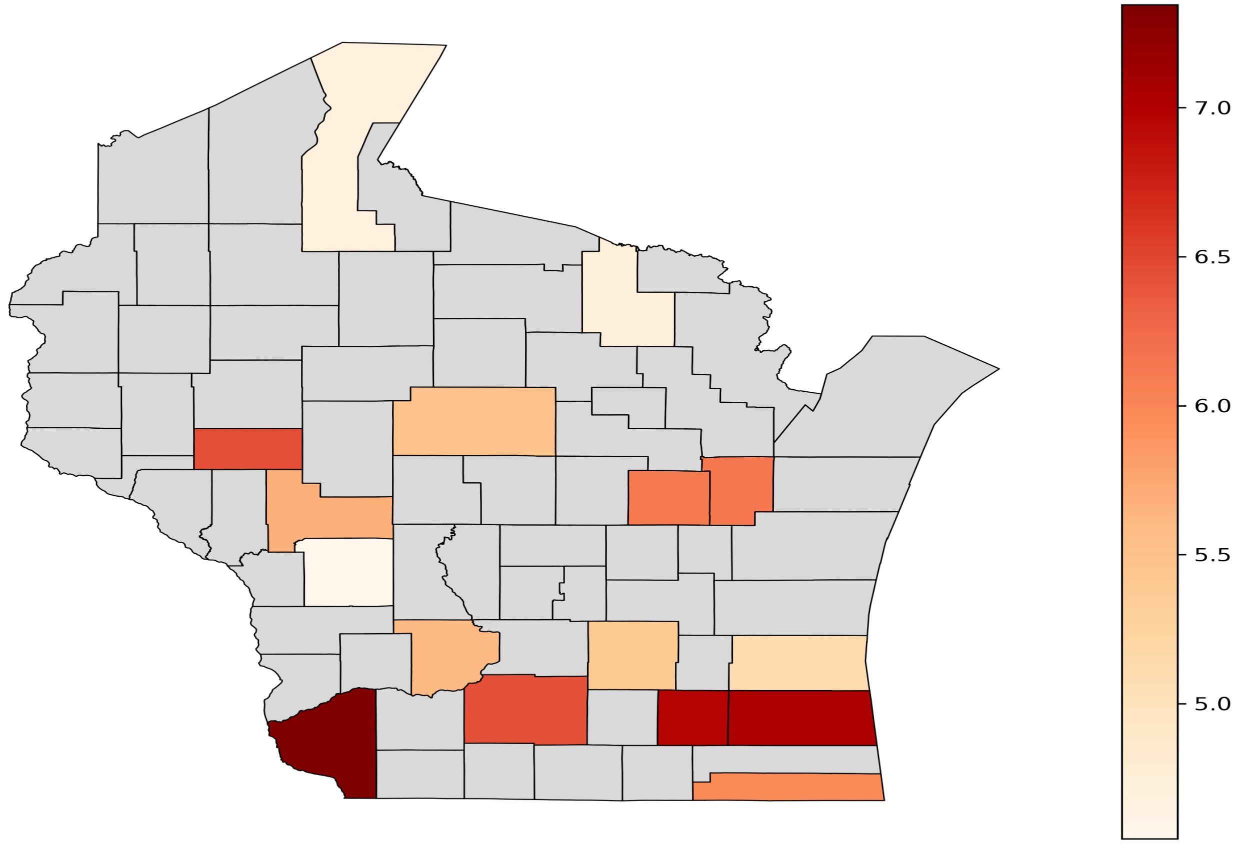
Limitations

This remains a monitor-based county summary rather than a statewide exposure surface. Sparse monitoring coverage, siting effects, and missing raw daily observations prevent stronger claims.

Policy relevance and future work

The project is most useful as a transparent demonstration of reproducible geospatial workflow, careful source documentation, and honest communication of spatial-statistical uncertainty.

Wisconsin County-Level Annual Mean PM2.5 (2024)



Monitored Wisconsin Counties Ranked by Annual Mean PM2.5